

Week 4-5, October 2 and 4, 2024

Main topics

- Tikhonov regularization
- Choice of Regularization Parameter
- Homework 1
- Computerized Tomography: Beyond filtered backprojection

Material

From last week: Kirsch 2.2. Pay attention to the equivalence of (2.15), (2.16) and (2.19).

New material this week: P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, Chapter 5 (skip NCP analysis 5.5).

Program October 2

13:00-13:30 Tikhonov regularization and matlab.

13:30-15:30 Exercise session - Tikhonov regularization. Matlab experiments, filter factors.

15:45-17:00 Lecture: Choice of Regularization Parameter

Program October 4

12:00-14:00 Homework 1

14:00-15:00 Research lecture on “Computerized Tomography: Beyond filtered backprojection” by senior researcher Jakob Sauer Jørgensen, DTU Compute.

15:00-16:00 Homework 1 cont.

Homework

The problems for homework 1 will be distributed on Wednesday October 2. You can then work with the problems during the week and in the session on October 4. The deadline for submission of solutions is set to Friday October 11. It is a requirement for attending the exam and passing the course that all homeworks are approved.

Exercises

You are highly encouraged to discuss and solve the exercises in small groups. If you are not very comfortable with infinite dimensional analysis (functional analysis) then you can think of a matrix $A \in \mathbb{R}^{n \times m}$ and vector $y \in \mathbb{R}^n$ whenever you encounter K between Hilbert spaces.

1. For TSVD and Tikhonov regularization the filter factors are given by

$$q(\mu) = \begin{cases} 1 & \mu > \alpha^2 \\ 0 & \mu \leq \alpha^2 \end{cases}, \quad q(\mu) = \frac{\mu^2}{\mu^2 + \alpha}.$$

Plot the filter factors (in loglog) for both TSVD and Tikhonov regularization for different truncations and regularization parameters α . Discuss pros and cons for using one or the other.

2. By calculating the 2-norm, show that for a matrix $A \in \mathbb{R}^{m \times n}$, vector $y \in \mathbb{R}^m$ and $\alpha > 0$

$$\|Ax - y\|_2^2 + \alpha \|x\|_2^2 = \left\| \begin{bmatrix} A \\ \sqrt{\alpha}I \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2,$$

with $I \in \mathbb{R}^{n \times n}$ denoting the identity matrix and $0 \in \mathbb{R}^n$. Conclude that the Tikhonov solution can be obtained by minimizing

$$J_\alpha(x) = \left\| \begin{bmatrix} A \\ \sqrt{\alpha}I \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2.$$

This is a typical least squares problem that can be solved efficiently e.g. in Matlab using `backslash(Aalpha \ b)`.

3. Show that for a compact operator K between Hilbert spaces X and Y and $\alpha > 0$, $(K^*K + \alpha I)$ is invertible:

- Show first that the operator is injective.
- If you are not comfortable with functional analysis, consider $K = A \in \mathbb{R}^{m \times n}$ and draw the conclusion from linear algebra.
- If you are familiar with functional analysis, invoke the Fredholm alternative.

4. Consider the Abel model problem and generate for $n = 100$ as in previous weeks the matrix A , the sampled function x , the righthand side $y = Ax$ and noisy $y^\delta = y + n$, with relative noise level δ .

- Fix the noise level $\delta = 0.01$ and compute (using the least squares formulation from exercise 1.) the Tikhonov regularized solution x_α for $\alpha = 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^0$. Plot the reconstructions and comment on the result.
- Repeat the above for $\delta = 0.05$. What is the optimal regularization parameter α (what does that mean)?

5. The compact operator $K : X \rightarrow Y$ has singular system $\{x_j\}, \{y_j\}, \mu_j$. Use the fact that

$$(K^*K + \alpha I)x_j = (\mu_j^2 + \alpha)x_j \\ K^*y_j = \mu_j x_j$$

to derive the Tikhonov reconstruction formula

$$x_\alpha = \sum_{j=1} \frac{\mu_j}{\mu_j^2 + \alpha} (y_\delta, y_j) x_j.$$

6. For the backwards heat problem ($T = 0.1$), create for $M = 100$ the matrix. Illustrate in the same plot (loglog) the svd factors ($\mu_j/q(\mu_j)$) for Tikhonov ($\alpha = 10^{-4}$), TSVD ($k = 5$). Along the first axis, put $\log \mu_j$ and out the second axis, the filter factors.
7. For the Phillips model problem, generate the matrix A ($n = 100$) and vectors x and $b = Ax$. Add noise (relative noise level $\delta = 0.05$.) Find for $\alpha = 10^{-8}, 10^{-7}, \dots, 10^{-1}$ the Tikhonov solution x_α . Investigate the Tikhonov functional J_α by plotting (loglog) the residual/misfit $\|Ax_\alpha - b\|_2$ (first axis) and solution norm $\|x_\alpha\|_2$ (second axis). What is the optimal balance between the two terms?
8. Show that the Tikhonov functional

$$J_\alpha(x) = \|Kx - y^\delta\|^2 + \alpha\|x\|^2$$

is strictly convex. Use this fact to prove uniqueness of a minimizer of the functional.